

<https://doi.org/10.1038/s41746-024-01349-5>

A virtual scalable model of the Hepatic Lobule for acetaminophen hepatotoxicity prediction



Stelian Camara Dit Pinto¹, Jalal Cherkaoui^{1,2}, Debarshi Ghosh¹, Valentine Cazaubon¹, Kenza E. Benzeroual³, Steven M. Levine⁴, Mohammed Cherkaoui¹, Gagan K. Sood⁵, Sharmila Anandasabapathy⁵, Sadhna Dhingra⁶, John M. Vierling⁷ & Nicolas R. Gallo^{1,3} ✉

Addressing drug-induced liver injury is crucial in drug development, often causing Phase III trial failures and market withdrawals. Traditional animal models fail to predict human liver toxicity accurately. Virtual twins of human organs present a promising solution. We introduce the Virtual Hepatic Lobule, a foundational element of the Living Liver, a multi-scale liver virtual twin. This model integrates blood flow dynamics and an acetaminophen-induced injury model to predict hepatocyte injury patterns specific to patients. By incorporating metabolic zonation, our predictions align with clinical zonal hepatotoxicity observations. This methodology advances the development of a human liver virtual twin, aiding in the prediction and validation of drug-induced liver injuries.

Drug-induced liver injury (DILI) is defined by injury to hepatic tissue resulting from a toxic reaction from a xenobiotic. In the USA and EU, the clinical burden is massive with DILI being the most common cause of acute liver failure (ALF)¹ potentially leading to death². DILI caused by APAP and idiosyncratic DILI accounts for more than 50% of all cases of ALF in the USA³. DILI is also major cause of new drug failure during clinical development and post-approval. A 2011 study showed that DILI was responsible for 33% of drug discontinuation and, hence, a major contributor to the high cost of drug development⁴. A 2011 analysis of pharmacovigilance data in phase IV (post-market evaluation) attributed 14.8% of withdrawals to hepatotoxicity, which was second only to cardiotoxicity⁵. Since then, the US Food & Drug Administration (FDA) has reported that DILI has been the most frequent, single cause of safety-related drug marketing withdrawals for the past 50 years⁶. In 2015, a study by the pharmaceutical industry (AstraZeneca, Eli Lilly, GlaxoSmithKline, and Pfizer) exploring the attrition rate in new drug development concluded that safety and toxicology were the largest contributing factors to drug attrition⁷. The cost of novel therapeutic development has increased steadily in the past decades (>1.7 billion 2011 USD⁸), while the success rate of bringing novel drugs to market has remained constant⁹. Other studies a decade later in 2021 found that R&D costs from 2011 onwards continued to increase¹⁰ and that DILI is still a leading cause of attrition during drug development¹¹. As of 2024, this unsolved problem remains the focus of intense research efforts to mitigate

both the clinical consequences of DILI and its negative impact on drug development^{12,13}.

Hepatotoxicity as a cause of new drug attrition contributes substantially to increasing R&D costs and calls attention to the unmet need to develop novel methodologies to address and predict DILI in humans early in the drug development process. Recent studies have increasingly highlighted the limitations of animal models in accurately predicting DILI in humans^{14,15}. Liver injury at extremely high doses in animal models has been shown to not necessarily predict direct hepatotoxicity in humans¹⁶. Another study showed that 43% of clinical toxicities were not predicted from animal studies¹⁷. Concordance between preclinical models and human clinical trials was seen in 63% of non-rodent studies, primarily in dog, and 43% in rodent studies, primarily in rat. The average concordance between liver toxicity in humans and animals models is 55%¹⁸. One study noted that animals studies failed to predict 67% of hepatotoxicity cases in clinical trials that necessitated termination of drug development¹⁹. Therefore, the need for more precise, non-invasive tools to better evaluate pre- and post-market hepatotoxicity in humans is great. Here, we showcase the initial development of a virtual model of the human liver lobule: a first, and necessary step in developing a multi-physics scalable digital platform to capture physical and biological processes of drug metabolism.

Identifying and addressing hepatotoxicity late in the drug development process in phase III trials or post-marketing surveillance in phase IV

¹Department of Computer Science, Digital Engineering and Artificial Intelligence, Long Island University, Brooklyn, NY, USA. ²Institut National des Sciences Appliquées, Lyon, France. ³Arnold & Marie Schwartz College of Pharmacy and Health Sciences, Long Island University, Brooklyn, NY, USA. ⁴Dassault Systèmes, Vélizy-Villacoublay, France. ⁵Division of Gastroenterology, Baylor College of Medicine, Houston, TX, USA. ⁶Department of Pathology and Genomic Medicine, Houston Methodist Hospital, Houston, TX, USA. ⁷Departments of Medicine and Surgery, Baylor College of Medicine, Houston, TX, USA.

✉ e-mail: nicolas.gallo@liu.edu

significantly raises costs and delays the availability of many potentially life-saving drugs. This prolongation of the development process, spanning years, results in direct cost increases for pharmaceutical companies and indirect costs for patients. Such delays have the direct effect of deferring or denying the availability of otherwise effective drugs to patients, while also adding a substantial financial burden that must be born by the cost of those drugs that do succeed. In 2011, Watkins identified the need to develop novel methodologies to address these issues⁵, specifically the need for novel methods providing insight in drug toxicity in humans early in the drug development pipeline. Toward this need, we propose creating a human liver virtual twin (VT) or a “Living Liver” to study drug attrition and accelerate drug development. The Living Liver could allow for the collaborative integration of multiple disciplines exploring DILI and allow for more accurate predictions of potential hepatotoxic effects, discovery, and validation of reliable biomarkers. It will provide scientists, industrials, and regulators with a novel digital platform to inform novel regularity strategies and drug safety sciences⁵.

A VT of a human organ can be defined as a detailed, computational model that accurately replicates certain anatomical structure, physiological functions, and biological responses of the corresponding organ in a human body. These models integrate data from various sources, including imaging, genetics, and clinical records, to create a dynamic, personalized representation that can be used for simulation, diagnosis, treatment planning, and drug testing. In their most advanced use, VTs can enable individual clinical analysis and prediction of organ behavior under different conditions, providing a powerful tool for personalized medicine and the avoidance of undesired side effects such as hepatotoxicity.

As a screening tool for drug toxicity, biomedical VTs must intrinsically be multi-scale and multi-disciplinary. They can include some or all aspects relevant pathophysiological functions at the level of molecules, cells, tissues, organs, or populations²⁰. While the VT construct is well established in non-medical disciplines, Baillargeon et al. pioneered the first human heart VT in 2014: the Living Heart model (SIMULIA, Dassault Systèmes)²¹. The Living Heart is a multi-scale mechanistic VT of the human heart coupling the electrophysiological and structural mechanics of the human cardiac function. It serves as a template methodology for other organs to integrate data, test theories, and design new solutions for biomedical sciences. The approach integrates existing laboratory research methods and offers a unique potential for translation to human body phenomena. Encouraged by this achievement, multiple stakeholders have endorsed this approach including researchers, clinicians, and the FDA, and is rapidly becoming a technique of choice for preclinical and clinical research, already leading to significant advances in medical device design, tests^{22–24}, and surgical planning^{25–28}. Ongoing efforts to build human organs VTs include the Brain^{29,30}, Kidney³¹, and Lung^{32–34}, suggesting multi-organ models may be possible.

The development of the liver VT has lagged behind other organs largely due to challenges in accurately modeling the liver’s functional unit, the lobule^{35,36}. While significant progress has been made at modeling the molecular, cellular, and macro scale phenomena, there is a crucial gap in lobule modeling: the functional scale, of the order of 1 mm. In this work, we

propose the development of the Virtual Hepatic Lobule as the key building block for the liver VT. The Virtual Hepatic Lobule captures the combined blood flow from the hepatic arteriole and portal vein to the central vein. By coupling the development of the lobule with a pharmacokinetic model of acetaminophen (APAP) intralobular zonal injury at the cellular level, our approach will allow for spatial prediction of APAP-induced damage within lobules.

The human liver is a multi-segment organ located in the upper right quadrant of the abdomen (Fig. 1). It is responsible for conducting several key physiological functions: protein synthesis, bile production, immune factor production, metabolism, and detoxification. Physically, the liver is composed of two main lobes, left and right, while each lobe is made up of millions of cells. Hepatocytes, the main functional cells of the liver, are between 12 and 24 μm in size³⁷ and align along sinusoidal units to form the hexagonal structural and functional unit called the liver lobule³⁸, acting as a miniaturized representation of the entire organ³⁹.

Lobule in-vivo geometry varies spatially across the liver but the hexagonal shape with portal triads at each corner and the central vein in the center is a commonly accepted description^{40–44}. Lobules are considered to be around 1 mm in diameter with a portal vein and central vein of 50 and 80 μm , respectively in diameter^{40–44}. When multiple hexagonal lobules are joined together, each portal tract supplies arterial and portal venous blood to three adjacent lobules. The liver lobule exhibits three distinct regions: the periportal zone, the mid zone, and the pericentral zone. Each of these areas possesses specific metabolic functions, known as metabolic zonation, that are driven by the oxygen gradient⁴⁵. Mathematical modeling of metabolic zonation provides insights into DILI at the lobular level. This paper introduces a coupled model focused on the liver lobule’s metabolic zonation to predict spatial patterns of APAP-induced damage within the lobules.

Multiple models of blood flow in the lobule have been created. A porous material structure is often adopted for hepatic tissue^{41–43,46–48}. Rani et al.⁴⁹ and Sluka et al.⁵⁰ chose to focus on a sub-unit of the lobule called the sinusoid unit. They modeled the inlet, outlet, and sinusoids as fixed geometries. However, this approach does not enable us to understand how this flow behaves in between hepatocytes at the full lobule scale. Rezanian et al.^{37,51,52} proposed a 2D lobule structure composed of squared subsections of porous media and sinusoids surrounding it. They then reconstructed the lobule geometry as a lattice structure and captured the non-heterogeneity of the liver and its impact on drug distribution throughout a large macroscopic area resembling the liver. Coombe et al.⁴⁷ extended this method to build a 3D model of the lobule, and couple their lobule model with a vasculature of a human liver using dual continuum upscaling to explore paclitaxel transport and metabolism in the liver. Bonfiglio et al.⁴² used a 2D model of a hexagonal shape with isotropic porous media properties to represent the blood flow in the lobule. Debbaud et al.⁴¹ demonstrated the importance of using an anisotropic permeability model to represent lobular blood flow. A similar approach has been used also incorporating lymph production^{40,44}. Hu et al.⁴³ took it one step further to represent the effects of pathologies such as fibrosis and cirrhosis. Lambers et al.³⁸ developed a comprehensive in-silico model that encompasses a continuum biomechanical framework based on the theory of porous media. This model was designed for the numerical

Fig. 1 | Visualization of the different scales of the human liver. From left to right: the organ scale (cm), the liver; the functional scale (mm), the lobule, and the cellular scale (μm) with hepatocytes in beige, stellate cells in blue, Kupfer cells in orange and endothelial cells in pink.

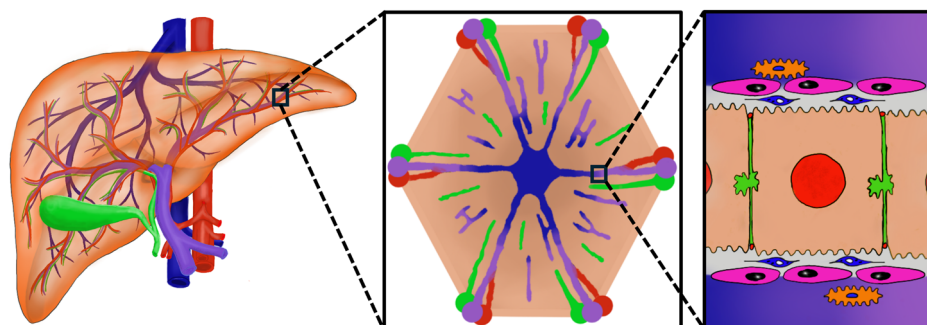
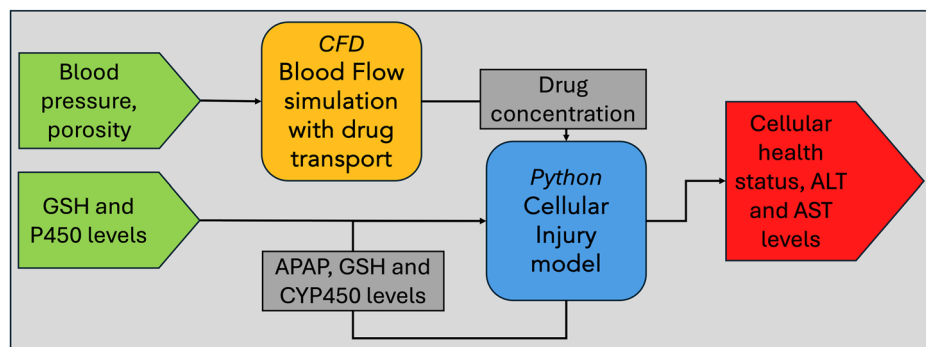


Fig. 2 | Virtual Hepatic Lobule APAP injury computational pipeline. Green blocks depict input to the pipeline. The orange block represents the CFD component. The blue block represents the cellular injury model. The gray boxes depict the data exchange between blocks. The red block represents the output of the computational pipeline.



simulation of interconnected perfusion–function–growth interactions within liver lobules in two dimensions. The developed model integrates spatially and temporally dependent liver perfusion, metabolic processes, and fat accumulation, reflecting their mutual influences in the metabolism of known hepatotoxic drugs.

N-acetyl-*p*-aminophenol (APAP, generic name acetaminophen) is a widely used “over-the-counter” analgesic-antipyretic medication with well-documented mechanisms of action. APAP hepatotoxicity is one of the most common, and unpredictable cause of DILI^{53,54}. It can result in ALF requiring liver transplantation^{55–59}. With liver transplantation being the second most common solid organ transplant and having only 10% of current liver transplant needs met⁶⁰, the need to address this clinical burden is great. In 2000, Shakil et al.⁵⁷ reported that 28% of transplant cases were attributed to APAP injury during a 13-year retrospective study from the University of Pittsburgh. Another study found in 2003 that APAP-induced ALF accounted for 20% of liver transplant cases⁶¹. The breadth of available data makes APAP the ideal agent to validate our approach. After ingestion, APAP reaches hepatocytes through the portal vein and is metabolized through phase II glucuronidation and sulfation enzymes⁶². About 5–10% of APAP is metabolized via phase I enzymes, into toxic *N*-acetyl-*p*-benzoquinone-imine (NAPQI)⁶². Glutathione (GSH) binding to NAPQI forms the nontoxic conjugate, mercapturic acid, which can then be excreted from the body as urine⁶². Toxic NAPQI occurs when hepatic stores of sulfate, glucuronide, and GSH are depleted or when phase I enzymes have been induced. Case in point, ingestion of excessive APAP can deplete the levels of GSH required to detoxify NAPQI. Toxic NAPQI is then free to bind to intracellular organelles causing oxidative stress that can injure or kill hepatocytes. The resulting hepatocellular necrosis can precipitate life-threatening ALF⁶².

Given the zonal nature of the liver, transport dynamics of APAP may play an important role in the hepatotoxicity process. The diffusion of APAP to the hepatocytes through the blood has been studied from varying perspectives. Using a fluid dynamics approach, Banaeiyan et al.⁶³ studied a convection–diffusion model for the diffusion of glucose in liver-lobule-on-a-chip. Diaz et al.⁶⁴ proposed the use of an advection–diffusion model to describe the distribution of APAP in the blood. This model described the evolution of the drug concentration in sinusoids based on the fluid velocity and the drug diffusion coefficient.

Similarly, multiple models using different mathematical foundations have been defined to simulate and predict the impact of drugs on the hepatocytes. Commercial software such as DILIsym® has been developed to predict potential drug hazards leveraging quantitative system pharmacology^{65–67}. However, this approach does not consider spatial heterogeneity of damage within the lobule or the liver. Damian and Raabe⁶⁸ proposed a model based on experimental data that could correlate in-vitro and in-vivo results. This model does not simulate the evolution of the drug effects directly, it suggests a first step to enable the use of in-vitro data to extrapolate in-vivo results. Edginton and Willmann⁶⁹ proposed to modify the whole-body physiologically based pharmacokinetic (WB-PBPK) model to consider the effects of liver cirrhosis. This work showcases the adaptability

of mathematical models to represent specific conditions. However, it does not consider the specifics of APAP toxicity nor output of results at the whole-body level. Ben-Shachar et al.⁷⁰ defined a mathematical model specific for APAP-induced hepatotoxicity to model the impact of APAP in the gut, plasma, liver, tissue, and urine. This complete model, composed of 21 ordinary differential equations, is used in combination with a model of GSH metabolism to provide results regarding the evolution of the patient and its possible recovery. This model is very complete yet focuses on a whole-body approach that is not specific to hepatocytes’ condition and necrosis.

Remien et al.⁷¹ proposed a liver toxicity model focused on hepatocyte damage at the liver scale. This model describes the evolution of APAP based on the concentration of GSH, and the concentration of alanine aminotransferase (ALT) and aspartate aminotransferase (AST) generated by cell injuries in a lobule. This model also describes the necrosis of hepatocytes based on the level of GSH and cytochrome P450 (CYP450) enzymes in hepatocytes. This model was used to match patients’ AST and ALT levels to simulated values to identify the dosage and the time of the overdose. Ghosh et al.⁷² leveraged their model as a base to explore the effects of alcohol consumption on APAP-induced toxicity. Both models were defined for description of macro scale effects but can be scaled down and adapted to a cellular scale. Moreover, since this model describes the necrosis of hepatocytes based on the level of GSH and CYP450 enzyme in the cell, the model can be modified to capture the zonation phenomenon⁷³. Recently, Gerhauser et al.⁷⁴ developed digital twin model of liver employing porous media to describe transport, mechanical properties, and deformations at the lobular level. Emphasis was placed on hepatic metabolism and damage, with simulations focusing on perfusion–zonation–function relationships. This model investigates the impact of zonation patterns and drug concentration on spatially heterogeneous liver damage, akin to drug toxicity models.

In this study, we present the development of the Virtual Hepatic Lobule, a dynamic, mechanistic predictive model that integrates blood flow dynamics and zonation effects to predict APAP-induced liver injury at the cellular level. The model captures spatial patterns of APAP-induced damage within the liver lobule by incorporating metabolic zonation. Through this multi-physics approach, we demonstrate the ability to predict hepatocyte damage based on pharmacokinetics and intralobular variability. Our results align with clinical observations of zonal hepatotoxicity, offering a novel scalable platform for simulating DILI and supporting broader toxicology applications.

Methods

APAP overdose clinical data

For our study, selected patients’ data came from the study by Remien et al.⁷¹ in 2012 where they reported a Model for Acetaminophen-induced Liver Damage (MALD) predicting the time of APAP overdose based on clinical data of patients presenting to emergency rooms. Data from 52 anonymized patients from Remien et al.⁷¹ with key clinical data for our approach: AST and ALT levels, time since ingestion, and initial APAP quantity. We selected four patients out of the prospective datasets to illustrate our approach. Those patients have been selected based on the quantity of APAP ingested to

Fig. 3 | Hexagonal representation of the human lobule and zonation. **a** Geometry of the lobule model depicting the zone boundaries. The red circle represents the portal veins (pressure inlet). The blue circle represents the central vein (pressure outlet). Three shades of beige depict the three lobular zones 1 through 3. **b** Depiction of metabolic zonation from the portal vein to the central vein driven by the oxygen gradient.

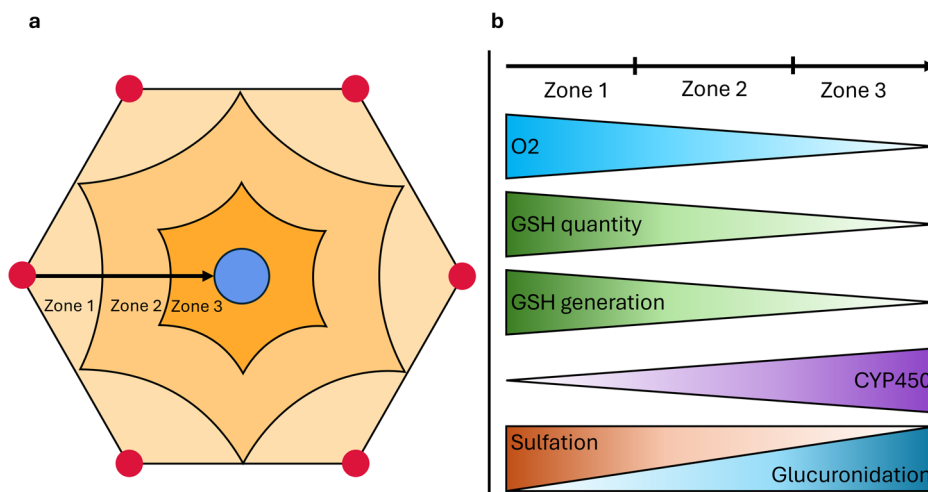


Table 1 | Blood flow and drug transport model parameters

Parameter	Value	Units	Reference
Lobule diameter	1.5	mm	40
Central vein diameter	0.15	mm	44
Portal track diameter	50	μm	44
Hepatocytes diameter	17.5	μm	37
Pressure outlet	500	Pa	43
Pressure inlet	800	Pa	43
Void ratio	0.143	–	43
Permeability	1.56e – 14	m ²	43
Blood density	1050	kg/m ³	43
Blood viscosity	3.50e – 05	kg/(m s)	43
Albumin molecular weight	66,500	kg/kmol	74
APAP density	1260	kg/m ³	75
APAP molecular weight	151.163	kg/kmol	75
Mixture mass diffusivity	2.22E – 10	m ² /s	76

represent low overdose, overdose, medium overdose, large overdose scenarios. Patient A ingested 6.1 g of APAP 3.9 days before its admission to the hospital and shows low damage. Patient B ingested 10.5 g of APAP 4.7 days before admission to the hospital and showed mild damage. Patient C ingested 20.5 g of APAP 3.4 days before its admission to the hospital and showed medium damage. Patient D ingested 27.1 g of APAP 4.5 days before its admission to the hospital and showed severe damage.

Ethics statement

Patient consent was not obtained because data were retrospective, were based on standard care, and were analyzed anonymously⁷¹. The protocol was approved by the Institutional Review Board (IRB) of the University of Utah in accordance with the Declaration of Helsinki⁷¹.

Virtual Hepatic Lobule APAP injury pipeline

The Virtual Hepatic Lobule APAP injury pipeline comprises two components: (1) the computational fluid dynamic (CFD) of the blood flow coupled with the drug transport, and (2) the cellular injury model (Fig. 2). These components are run iteratively to simulate the propagation of the injury in the lobule based on the lobular blood flow. First, blood flow, with healthy properties, is simulated in the “blood flow simulation with drug transport” component. This porous flow simulation is coupled with a species transport and advection–diffusion model that provides us with the drug

concentration in the lobule. Then, the “cellular injury” model is applied to calculate the effect of the drug for each node of the model. The results from this injury model enable us to calculate the health status at every node of the mesh. This model uses simulations performed on ANSYS Fluent (version 24.1) coupled with a Python code (using Python 3.9). It is currently implemented with the use of APAP but will be generalized to perform analysis for multiple drugs and treatments. The details of each component are described below.

Drug diffusion in the blood

The model of the diffusion of the drug from the guts to the blood was defined based on the literature⁷⁵. This diffusion model is defined by crucial time-frames. At the moment of the ingestion of the drug by the patient (t), diffusion model is initiated. By $t + 1$ h, the diffusion starts. By $t + 3$ h the maximum concentration is attained, followed by the average concentration at $t + 6$ h. At the end of the process, at $t + 8$ h, the drug concentration is zero, no trace of drug is found in the blood.

Blood flow simulation with drug transport

The CFD blood flow simulation was performed on ANSYS Fluent (version 24.1) using a momentum conservation equation derived from Navier–Stokes equations including the description of the porous media effects (see Eq. 1).

$$\rho \left(\frac{\partial v}{\partial t} + v \cdot \nabla v \right) = -\nabla p + \nabla \cdot T + f \quad (1)$$

where v is the velocity, ρ the density, p the static pressure, T the stress tensor, and f the porous media source term.

The chosen geometry of the lobule model is the hexagonal 2D model presented in Fig. 3. The six blood inlets (portal veins) are set up as pressure inlets, and the blood outlet (central vein) is set up as a pressure outlet. The edges of the lobule are defined as symmetries to simulate the repeatability of the lobules in the liver. The lobule was defined as a porous material. This assumption reflects the complexity of sinusoidal blood flow running between hepatocyte cords in the lobule from the portal tracts to the central vein. This behavior is described by a void ratio (ratio between sinusoidal spaces containing flowing blood and hepatocyte cords separated from the sinusoids by the fenestrated liver sinusoidal endothelial cells [LSECs] and lymphatic Space of Disse) and a published permeability value^{41–43,46–48}. The dimensions selected for the model were defined based on the literature (see Table 1). The mesh size was set to 1.75e – 5m (average size of an hepatocyte) for a total of 5114 nodes.

The drug transport was simulated using the species transport model of the ANSYS Fluent solution. This solution allows the user to compute

chemical reactions, and it uses conservation principles for convection and diffusion to ensure the mass fraction of species at each node (see Eq. 2). To capture the diffusion phenomenon of APAP in the blood, the molecular weight of Albumin has been used. Albumin has been selected because it is the main constituent of the plasma in which APAP is diffusing⁷⁶.

$$\frac{\partial}{\partial t}(\rho Y) + \nabla \cdot (\rho \vec{v} Y) = -\nabla \cdot \vec{J} + R + S \quad (2)$$

Each simulation was performed using the transient, pressure-based solver. A fixed time step of 1 s ($\Delta t = 1$ s) was set to perform 1200 steps. At the end of the simulation, the concentration of APAP was then exported into a text file to be read by the Python code. However, the porous media assumption does not adequately account for the fact that the flow of drug is restricted to the sinusoids and not the total model volume. Therefore, drug concentration within the sinusoidal space was calculated using the defined void ratio. This new concentration of APAP was then used as input to the cellular injury model.

Cellular injury model

While the original MALD model was designed to simulate the evolution of the toxicity effect of APAP throughout a full liver⁷¹, our model focuses on simulating the effects of the toxicity at the level of individual cell. Therefore, a scaled-down version of the MALD model has been developed.

The modified approach describes the evolution of the mass (Eq. 3), NAPQI mass (Eq. 4), and GSH mass (Eq. 5). The hepatocyte has four possible states: healthy (Eq. 6), damaged (Eq. 7), necrosed (Eq. 8), and regenerated (Eq. 9). The state of the hepatocyte is based on the quantity of NAPQI. Furthermore, the metabolized concentration of AST (see Eq. 10) and ALT (see Eq. 11) can be composed based on the state of the necrosis.

$$\frac{dA}{dt} = \alpha * A - \delta_A * A \quad (3)$$

$$\frac{dN}{dt} = p * \alpha * A - \gamma * N * G - \delta_N * N \quad (4)$$

$$\frac{dG}{dt} = \kappa - \gamma * N * G - \delta_G * G \quad (5)$$

$$\frac{dH}{dt} = r * H * (1 - H + Z) - \eta * N * H \quad (6)$$

$$\frac{dZ}{dt} = \eta * N * H - \delta_Z * Z \quad (7)$$

$$\frac{dL}{dt} = \delta_Z * Z - r * H * (1 - H + Z) \quad (8)$$

$$\frac{dR}{dt} = r * H * (1 - H + Z) \quad (9)$$

$$\frac{dS}{dt} = \frac{\delta_Z * \beta_S * Z}{\theta} - \delta_S * (S - S_{\min}) \quad (10)$$

$$\frac{dL}{dt} = \frac{\delta_Z * \beta_L * Z}{\theta} - \delta_L * (L - L_{\min}) \quad (11)$$

Our modified version of MALD shows a number of differences. First, the $H_{\max}$ value describing the total number of hepatocytes, $H_{\max}$, in the different equations has been omitted. Second, APAP is now treated constantly, without considering the number of healthy hepatocytes. Third, the conversion factor, q , from grams of APAP to moles of NAPQI has been removed. Lastly, Eq. (4) integrates a term accounting for NAPQI decay. The equations governing hepatocyte evolution have been adjusted to express the health status from 1 to 0, while hepatocyte necrosis and regeneration have

Table 2 | Model equations parameters converted from days to seconds and from the total number of hepatocytes to one based on human clinical data from patients with APAP hepatotoxicity⁷¹

Variable	Symbol	Value	Units
Amount of blood in human body	θ	5.0	L
Number of hepatocytes in liver	$H_{\max}$	1.6e11	cells
Damaged hepatocytes necrosis rate	δ_Z	5.79e − 5	/s
Hepatocytes regeneration rate	r	1.85e6	/s
Hepatocytes damage rate	η	5.94e8	cell/mol/s
APAP clearing rate	α	7.29e − 5	/s
APAP unconjugated clearing rate	δ_A	3.82e − 6	/s
APAP fraction oxidized	ρ	0.05	–
GSH bind to NAPQI rate	γ	1.85e13	cell/mol/s
GSH decay rate	δ_G	2.31e − 5	/s
GSH production rate	κ	1.59e − 19	mol/cell/s
NAPQI decay rate	δ_N	1.16e − 9	/s
AST clearing rate	δ_S	1.06e − 5	/s
ALT clearing rate	δ_L	4.05e − 6	/s
Total quantity of AST	β_S	200,000	IU
Total quantity of ALT	β_L	84,800	IU
Minimum AST	$S_{\min}$	12.0	IU/L
Minimum ALT	$L_{\min}$	9.0	IU/L

been isolated into their own equations. Moreover, equation parameters have been scaled to seconds rather than days to match the time step used in the flow simulation (Table 2).

This model was implemented in Python 3.9 on Anaconda3 and Spyder 5.4.3 GUIs. The coupled ODEs were solved using the SciPy package and the Implicit Runge–Kutta method. This method was chosen due to its effectiveness in handling stiff equations, where other solvers were found to be numerically unstable.

Zonation model

Liver zonation refers to the organization of different metabolic activities within the liver into distinct zones or regions (Fig. 3). Different studies have explored the root cause of such spatial heterogeneity^{38,45,77–80}. This phenomenon takes place at the lobule scale and different zones have been defined to better represent its effects. The three zones of the lobule are depicted in Fig. 3. The first zone, or periportal zone, is the area closer to the portal veins. The second zone is the middle area equidistant from the portal and central vein. Finally, the third zone, or centra-lobular zone is the area closest to the central vein. When looking at hepatotoxicity, the zonation effect has been observed through microscopic imaging. Necrosis first appears in the centra-lobular zone and propagates toward the periportal zone⁵³. The exact reason for this effect is still unknown but multiple factors have been identified as driving this metabolic zonation such as an oxygen gradient⁴⁵, a nutrient gradient⁸⁰, the expression of Wnt/ β -catenin signaling⁸¹ or sinusoidal geometry induced phenomenon (fenestrations frequencies, porosity). A decreasing oxygen gradient causes hepatocytes of the centra-lobular zone to receive less oxygen than their counterpart in zone 1⁴⁵.

Although zonation is a well-observed phenomenon, to the best of our knowledge there were no previously published and validated findings on quantitative variations of different zonation parameters within the human lobule. Therefore, we define linear gradients where the spatial average over the lobule area matches reported values in the literature.

Below is described our zonation parametrization plan to capture metabolic heterogeneity. The first zonation implementation is a lower quantity of GSH present in the hepatocytes of the centra-lobular area. This was achieved by parametrizing a gradient of initial GSH levels in the

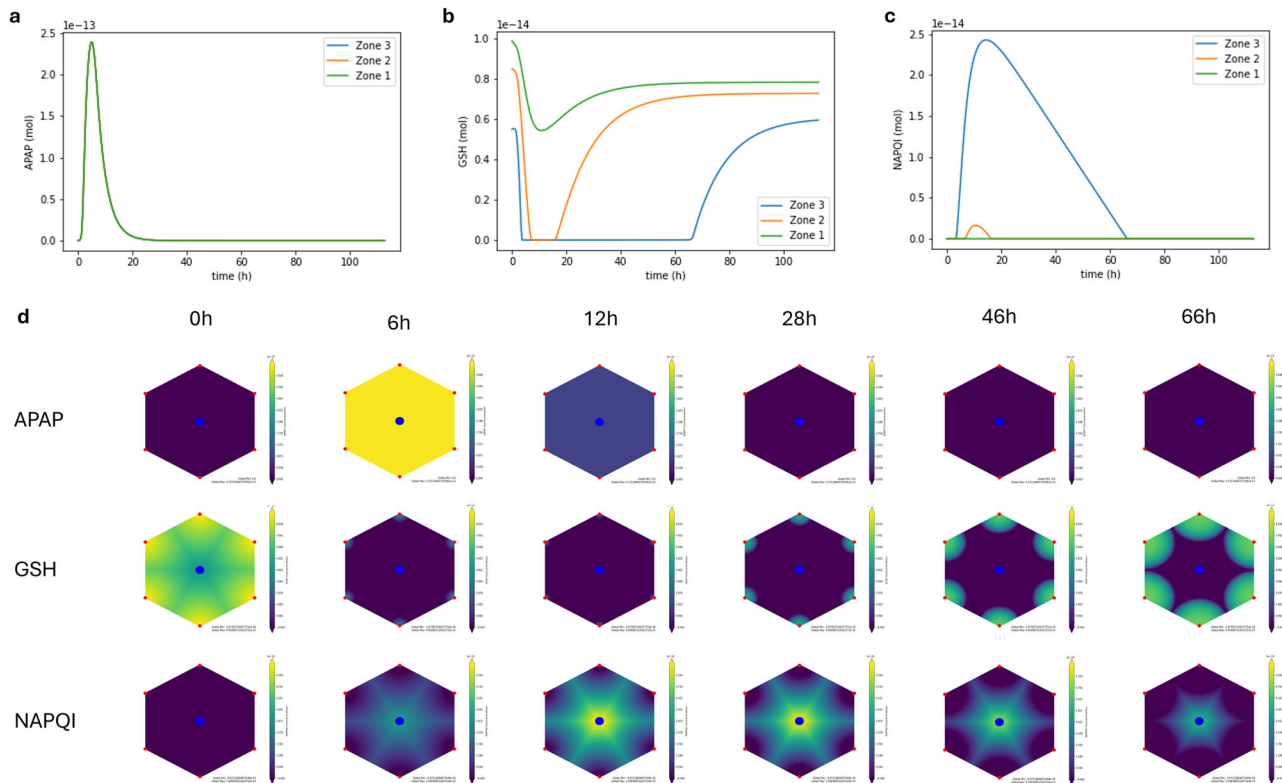


Fig. 4 | APAP, GSH, and NAPQI quantity evolution in the lobules for patient B. **a** Evolution of the quantity of APAP entering hepatocytes (mol). **b** Evolution of the quantity of GSH in hepatocytes (mol). **c** Evolution of the quantity of NAPQI

generated in hepatocytes (mol). **d** Heatmaps showing the quantity of APAP, GSH, and NAPQI in the whole lobule. Red circle represents the portal veins (pressure inlet). The blue circle represents the central vein (pressure outlet).

hepatocytes. The linear gradient is depicted in Eqs. (12)–(14) and ranges from $1\text{e} - 14$ mol in zone 1 down to $0.5\text{e} - 14$ mol in zone 3:

$$GSH_{cell} = a * r + b \quad (12)$$

$$a = \frac{GSH_{CV} - GSH_{PV}}{R_{Lobule}} \quad (13)$$

$$b = GSH_{PV} \quad (14)$$

The second zonation implementation captures the slower GSH regeneration (κ) in zone 3 (centra-lobular)⁴⁵ using a linear gradient ranging from $1.35\text{e} - 19$ mol in zone 1 down to $1.82\text{e} - 19$ mol in zone 3.

The third zonation implementation captures the difference in the binding of the NAPQI and GSH. The CYP1A2 enzyme is responsible for GSH to NAPQI binding and is found in greater amounts in zone 1^{2,45}. This implies that GSH binding to NAPQI is faster in zone 1, leaving less time for NAPQI to bind with the hepatocyte's membrane and trigger the necrosis process. We captured this phenomenon through a linear gradient of the GSH binding rate (γ) starting from $2.08\text{e}13$ mol in zone 1 down to $1.62\text{e}13$ mol in zone 3.

The fourth and last zonation implementation regards the oxidation of APAP into NAPQI, its toxic metabolite. This oxidation is due to the presence of more CYP2E1 enzymes in zone 3 compared to zone 1. Therefore, there is no APAP oxidation in zone 1, however, about 10% of APAP is oxidized in zone 3^{2,45}. We implemented this zonation of APAP oxidation through a linear gradient of the fraction of APAP being oxidized (p).

Statistical analysis

To validate the model, using the SciPy library in Python 3.9, we assessed damage across zones under both zonation and non-zonation scenarios. This assessment aims to discern the impact of zonation across various lobule

zones. Employing the χ^2 -test, tailored for analyzing categorical data like damage status, we accommodate the non-normal distribution inherent in damage data. Multiple datasets are utilized for this analysis. Initially, we analyzed the results for each zone to delineate the zonation's influence on individual zones per patient. Subsequently, we compared outcomes with and without zonation to highlight the zonation-induced effects across zones within the model. These analyses aim to illustrate the significant discrepancy between zones, consistent with the anticipated outcomes of the zonation model implementation.

Results

The drug diffusion model was developed based on the concentration previously described. In this diffusion model, the drug was diffused in the blood gradually. Additionally, the phenomenon of zonation of GSH was implemented in the model to improve its realism. To show the impact of the phenomenon on the variables, the evolution of APAP, NAPQI, and GSH in the portal (Zone 1), mid-lobule (Zone 2), and central region (Zone 3) are shown in Figs. 4 and 5 for patients B and C, respectively. We can observe across all three zones the evolution of APAP (Figs. 4a and 5a), GSH (Figs. 4b and 5b), and NAPQI (Figs. 4c and 5c). We can see that for both patients there is a decrease in GSH after APAP ingestion due to binding to NAPQI. GSH levels are depleted longer in zone 2 than in zone 1 and significantly longer in zone 3 compared to the other zones. GSH levels start recovering very fast in zone 1 compared to zone 3. These findings are expected based on the parametrization of zonation in our model. It is also worth noting that GSH levels were not recovered in either patient after 100 h. For both patients, the NAPQI level increases a few hours after APAP ingestion in zones 2 and 3 while it remains close to 0 in zone 1. As expected, it takes longer for NAPQI to be metabolized in Zones 3 and 2.

We can visualize the spatial heterogeneity of APAP metabolism through time since ingestion in Figs. 4d and 5d for patients B and C, respectively. One can appreciate the “star-like” pattern in GSH regeneration

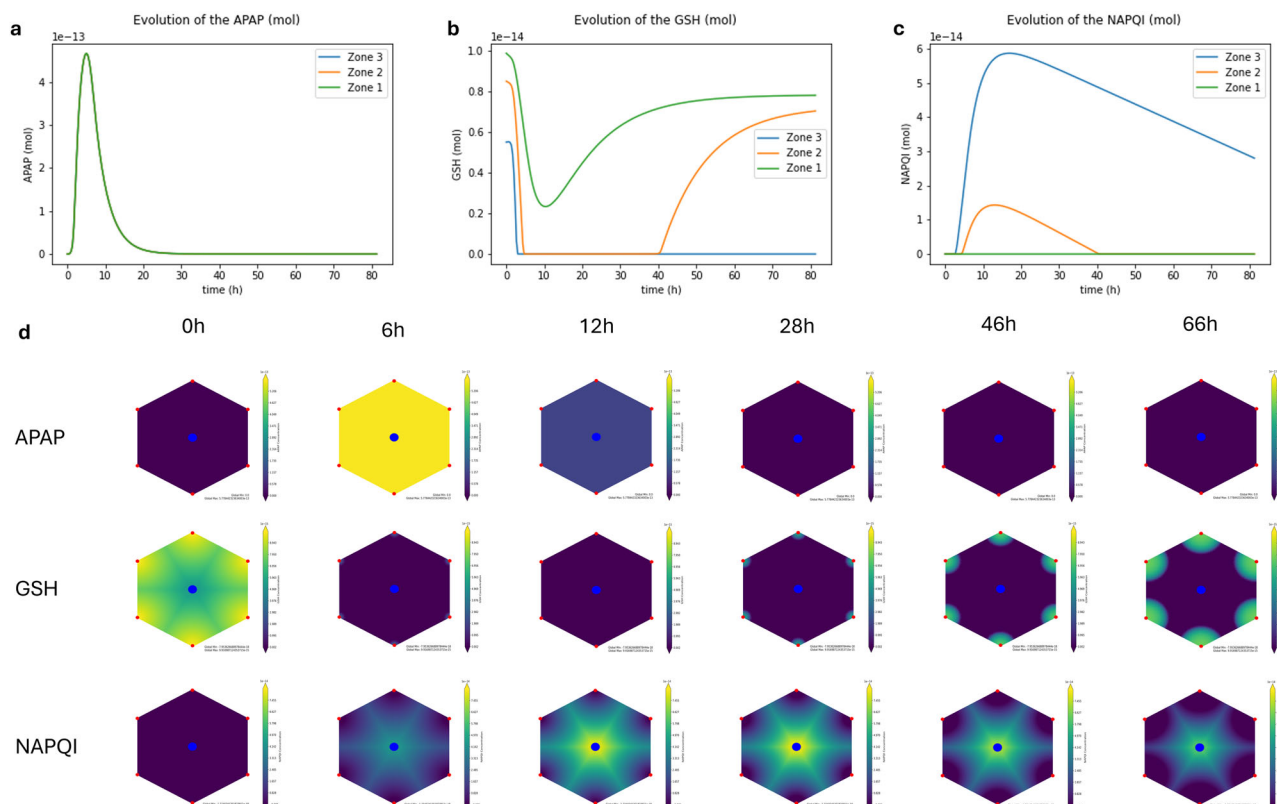
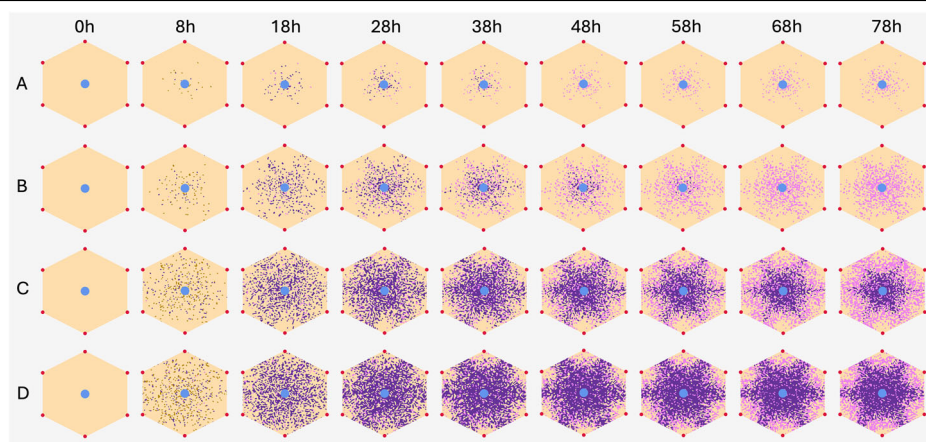


Fig. 5 | APAP, GSH, and NAPQI quantity evolution in the lobules for patient C. **a** Evolution of the quantity of APAP entering hepatocytes (mol). **b** Evolution of the quantity of GSH in hepatocytes (mol). **c** Evolution of the quantity of NAPQI

generated in hepatocytes (mol). **d** Heatmaps showing the quantity of APAP, GSH, and NAPQI in the whole lobule. Red circle represents the portal veins (pressure inlet). The blue circle represents the central vein (pressure outlet).

Fig. 6 | Lobular hepatocyte damage prediction maps over time for patients A, B, C, and D showing the hepatocyte health status (healthy in beige, damaged in brown, necrosed in purple, or regenerated in pink). Red circle represents the portal veins (pressure inlet). The blue circle represents the central vein (pressure outlet).



after 46 h and NAPQI metabolism after 28 h. GSH depletion happens promptly and homogenously throughout the lobule during the first 12 h. GSH regeneration then begins in zone 1 before the other zones. After 66 h GSH levels are still close to 0 in zone 3 and have not recovered to pre-APAP ingestion levels in zone 1 and 2 for both patients. NAPQI depletion begins in zone 3. After 66 h NAPQI is metabolized in zone 1 and halfway through being metabolized in zone 2. The quantity of NAPQI remains at the highest in zone 3.

Our approach enables visualization of hepatocyte damage prediction over time as either healthy, damaged, necrosed, or regenerated. These prediction maps for patients A, B, C, and D can be seen in Fig. 6. We can observe APAP metabolism and its impact on hepatocyte health status for all four

patients. Patients A, B, C, and D have, respectively, consumed 6.1, 10.5, 20.5, and 27.1 g of APAP. As one might expect, the rate of damage and number of injured or necrosed hepatocytes increases with the APAP quantity ingested. When observing the spatial heterogeneity of the damage in all four patients, we can observe more damage and necrosis in the centrilobular zone (zone 3) reflecting the well-documented star pattern of zonation (Fig. 6).

To ascertain the effect of the zonation, the hepatocytes in a specific zone were compared using both no-zonation and zonation models at several time points after APAP ingestion. At 18 h after ingestion, a significant difference was observed between the zonation model and no-zonation model for patient B in zone 1 (Fig. 7a, $p < 0.001$), zone 2 (Fig. 7b, $p < 0.05$), and zone 3 (Fig. 7c, $p < 0.001$). At 58 h after ingestion, a significant difference was

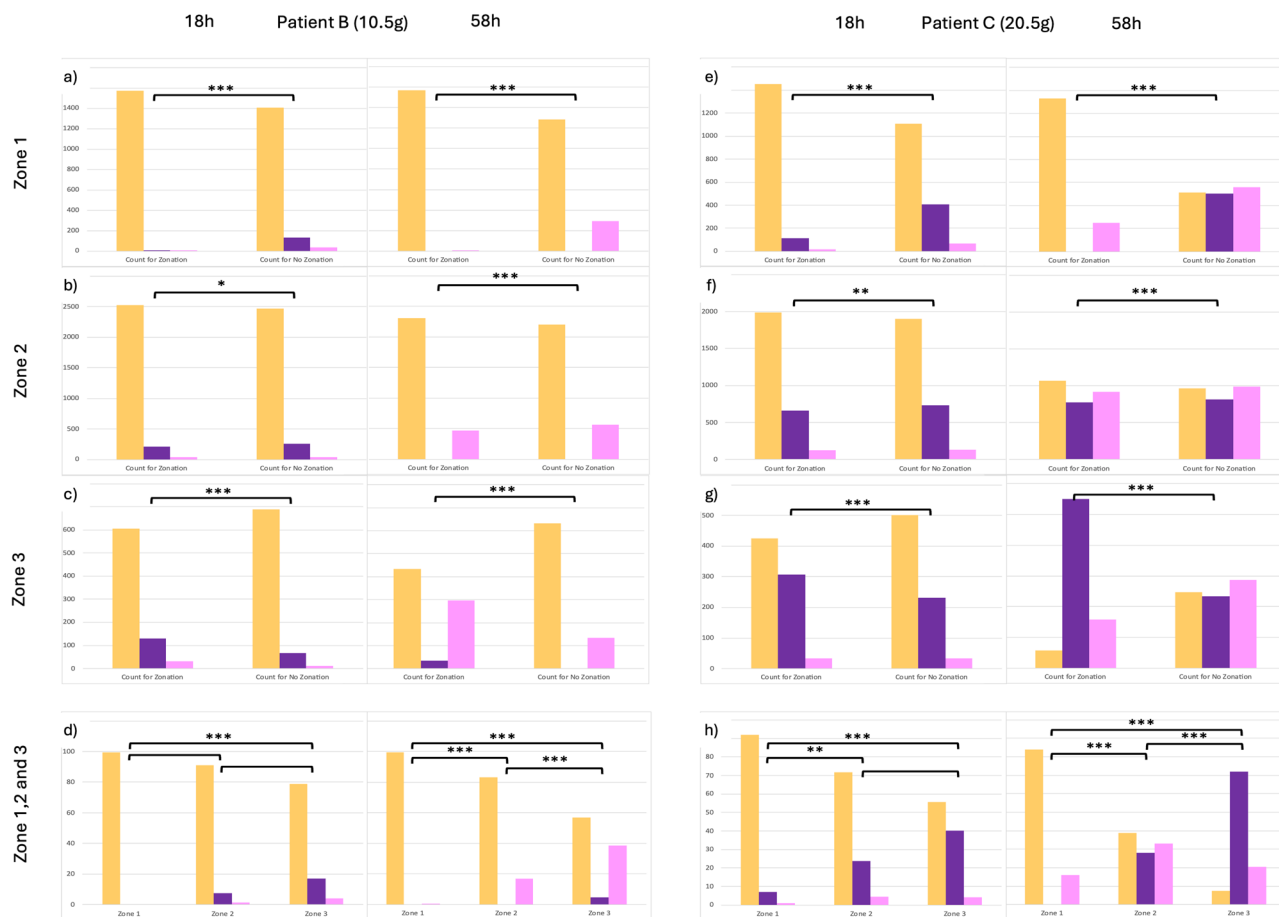


Fig. 7 | Statistical comparisons between zonation and no-zonation model for patients B and D. Specifically, comparing hepatocyte health across each zone at 18 and 58 h after APAP ingestion (healthy in beige, damaged in brown, necrosed in

purple, or regenerated in pink). **a** Zone 1, **b** Zone 2, **c** Zone 3, and **d** all zones for patient B. **e** Zone 1, **f** Zone 2, **g** Zone 3, and **h** all zones for patient D. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

observed between the zonation model and no-zonation model for patient B in zone 1 (Fig. 7a, $p < 0.001$), zone 2 (Fig. 7b, $p < 0.001$), and zone 3 (Fig. 7c, $p < 0.001$). Similarly, at 18 h after ingestion, a significant difference was observed between the zonation model and no-zonation model for patient C in zone 1 (Fig. 7e, $p < 0.001$), zone 2 (Fig. 7f, $p < 0.01$) and zone 3 (Fig. 7g, $p < 0.001$). At 58 h after ingestion, a significant difference was observed between the zonation model and no-zonation model for patient C in zone 1 (Fig. 7e, $p < 0.001$), zone 2 (Fig. 7f, $p < 0.001$), and zone 3 (Fig. 7g, $p < 0.001$).

To further explore the significance of incorporating zonation effects we compared the proportion of hepatocytes between zones within the zonation model. At 18 h after ingestion, a significant difference was observed for patient B between zones 1 and 3 (Fig. 7d, $p < 0.001$). However, no significance was observed for patient B between zone 1 and zone 2, and zone 2 and zone 3 (Fig. 7d, $p > 0.05$). At 58 h after ingestion, a significant difference was observed for patient B between zone 1 and zone 2, zone 2 and zone 3, and zone 1 and 3 (Fig. 7d, $p < 0.001$). Similarly, at 18 h after ingestion, a significant difference was observed for patient C between zones 1 and 2 (Fig. 7h, $p < 0.005$) and zones 1 and 3 (Fig. 7h, $p < 0.001$). At 58 h after ingestion, a significant difference was observed for patient C between zone 1 and zone 2, zone 2 and zone 3, and zone 1 and 3 (Fig. 7h, $p < 0.001$). It can be noted that the significance of the difference in hepatocytes' status grows over time (Fig. 7d, h). These results show that the parametrization of zonation within the hepatic virtual lobule significantly impacts spatial heterogeneity damage prediction at the lobule scale. These findings open an interesting research direction to estimate patient-specific zonation parameters from patient-specific quantities of GSH and CYP450 enzyme activity. This approach could enhance our understanding of the mechanisms behind spatial heterogeneity in DILI and improve personalized predictions of liver toxicity.

Discussion

The Virtual Hepatic Lobule model reported here combines fluid dynamics and pharmacokinetic models and demonstrates the capacity of the model to predict lobular zonal toxicity of APAP. Previous works on the subject have used different approaches. Fu et al.⁸² used a physiologically based pharmacokinetic (PBPK) model to describe the transport and metabolism, of xenobiotics in the lobule. Their work showed similar velocity profiles as the ones observed using the porous media fluid dynamics simulations in our work as well as in the literature^{40,42,43}. The literature showed that this model could be made more realistic by taking into consideration the difference in inlets between the hepatic artery and the portal vein⁴⁰ or even the presence of the vascular septa that connect the portal veins⁸³. Both implementations would ensure a more realistic blood flow pattern and hence a more realistic drug diffusion in the hepatocytes.

In addition to the modeling of the blood flow, the model of Fu et al.⁸², similar to the one used by Sluka et al.⁵⁰, describes the transport of APAP to the hepatocytes in the sinusoid unit, a sub-scale of the lobule. Both those works show a greater difference in APAP assimilation by hepatocytes along sinusoids than the one observed in the Virtual Hepatic Lobule model. This spatial difference in drug delivery was not considered in the current model. This effect was judged to be insignificant at the lobular scale due to the velocity at which blood diffuses inside the lobule. In addition, our model currently implements symmetric boundary conditions to account for neighboring lobules. When seeking to couple the Virtual Hepatic Lobule with a macroscopic liver vasculature to move toward a fully multi-scale liver model; one will have to consider spatial heterogeneity in lobule geometries. This will be important as cellular necrosis typically initiates inflammatory processes in neighboring cells, potentially triggering necrosis in adjacent

lobules⁶². Similarly, the model developed by Lambers et al.³⁸ analyzed fat metabolism in a lobule model by interconnecting perfusion–function and growth interactions within liver lobules. The model analyzed fat metabolic distribution and zonation by considering the effect of blood pressure as a boundary condition for different arrangements of lobules. Our Virtual Hepatic Lobule model currently assumes uniform static pressure and does not consider blood pressure as boundary conditions to account for lobule arrangements. The integration of fluid pressure as a boundary condition will be an important additional parameter when considering the spatial distribution of metabolites/drug in the neighboring cells or lobules.

Looking at the modeling of APAP-induced damages, the Virtual Hepatic Lobule model predicted centrilobular damage patterns that recapitulate the zonal hepatotoxicity observed in patients^{84,85}. Our hepatocyte damage prediction, which couples CFD and pharmacokinetics, matches damage prediction from quantitative systems toxicology (QST) modeling⁷³, cellular automaton (CA) based modeling⁶², and coupled porous media and cellular scale modeling⁷⁴. This multiple star pattern of the damage boundary matches pathology images from literature for APAP injury^{84,86}, further validating that the pharmacokinetic coupling was correctly adapted to the hepatocyte scale. Another way to predict APAP-induced damage is through the use of Liver-on-Chip (LoC). LoC have proven valuable as a substitute to animal model which sometimes fail to predict toxicity in humans due to interspecies differences in metabolic pathways⁸⁷. LoC have the capabilities to recreate liver micro-environment in-vitro and have been used to model drug metabolism^{88–90} and hepatotoxicity^{91–94}. However, LoC models are extremely complex to manufacture and scale. They are often designed for short- to medium-term experiments, focusing on localized cellular responses. We believe that LoC approaches and in-silico models, such as the Virtual Hepatic Lobule, are complementary as LoC can enable to calibrate in-silico models by exploring the depth of specific hepatocytes metabolism during DILI.

The zonation implemented in the Virtual Hepatic Lobule model follows the guidelines summarized by Kietzmann⁴⁵. Four phenomena were implemented in the Virtual Hepatic Lobule: (1) a gradient of GSH quantity, (2) a gradient of GSH regeneration, (3) a gradient of GSH and NAPQI binding, and (4) a gradient of APAP oxidation. However, the zonation phenomenon is still under study and genetic expression studies are helping to understand the phenomenon^{74,77–79,95}. The Virtual Hepatic Lobule is the ideal platform to explore, predict, and validate untested zonation hypothesis. It is also the ideal platform to couple other xenobiotics, pharmacokinetic, and metabolism models. For APAP, exploring a more detailed whole-body PBPK model such as the one proposed by Ben-Shachar⁷⁰ would grant a wider control over parameters controlling for zonation. Additionally, LoC provides knowledge by mimicking the complex micro-environment of the liver, including zonal variations (such as oxygen gradient⁹⁶) and cell–cell interactions. LoC models generate highly detailed experimental data that reflect real-time physiological responses. These empirical data can be used to validate and refine in-silico predictions, ensuring that the Virtual Hepatic Lobule accurately represents human liver function.

Additionally, other cellular phenomena can be added in future implementation to capture important cellular processes. Incorporating the immunological roles of natural killer (NK), natural killer T (NKT) cells, and activated macrophages after NAPQI-induced hepatocyte injury/death of action is one direction to explore to improve the performance of our approach⁷¹. In addition to the effect or functions of these cells, their known effects on coagulation in the sinusoid may also affect the blood flow. Such effects could be modeled by a change in porosity in the blood flow model. Such implementation would help provide a more accurate picture of the damage as time goes on.

In pharmaceutical development, unanticipated hepatotoxicity is estimated to be the cause of 22% of clinical trial failures and 32% of market withdrawals of novel molecular entities^{97–99}. As previously mentioned, animal studies have been shown to lack the predictive capacity of a new drug with hepatotoxic potential in humans. Clinically, hepatotoxicity remains the principal cause of ALF in the western world, with a case fatality rate ranging

from 10% to 50%⁶. The disease burden is significant considering that most DILI instances need liver transplantation. A human liver VT would provide an ideal digital platform to integrate areas of science previously siloed and integrate physiological models across all time and spatial scale of the liver. By nurturing a collaborative framework across multiple teams, the Living Liver seeks to build a multi-physics and multi-scale VT of the human liver to accelerate therapy development and reduce the clinical burden of DILI.

In this study, we sought to build the Virtual Hepatic Lobule, the building block and functional unit of the human liver VT and the first step toward the Living Liver virtual twin. By integrating blood flow dynamics and a clinically informed model of APAP overdose liver injury we demonstrated our capability to predict lobular zonal damage. Future research efforts and improvement include refining the cellular phenomenon captured within the Virtual Hepatic Lobule such as variability in basal concentrations of GSH; modeling the generation of new GSH resulting from intravenous treatment with *N*-acetylcysteine; modeling the variability of hepatocyte regeneration due to mitogens, nutrients and phosphate; modeling of prothrombin complex production in zone 3 regenerated hepatocytes (to normalize prothrombin time and international normalized ratio); and finally the modeling of the retrograde flow of hepatic lymph produced within the Space of Disse. Future efforts will also seek to incorporate new data about differential hepatic zonal gene expression involved in the transport and metabolism of xenobiotics, therefore allowing to parameterize our model with genetic information and move toward exploring the implications of pharmacogenomics on DILI. Future efforts will also focus in coupling the Virtual Hepatic Lobule with PKPD model of various xenobiotics to considering spatial heterogeneity and temporal-spatial distribution of absorption, distribution, metabolism, and excretion (ADME) dynamics at the organ level.

Based on these results, we conclude that expanding the methodology into a human liver VT would provide an ideal digital platform to address the high propensity of unanticipated hepatotoxicity in drug trials and precision drug dosage guidance to reduce the clinical burden of DILI. We demonstrate the ability to integrate areas of science previously siloed and physiological models across time and spatial scales. Further, a Living Liver VT could serve as a multi-disciplinary collaborative framework as demonstrated with the Virtual Hepatic Lobule.

Data availability

The data used to support the findings in this study are available at <https://doi.org/10.1002/hep.25656>.

Code availability

Python 3.9 code for APAP metabolism models and ANSYS Fluent source files for the coupled APAP diffusion and blood flow are available upon request.

Received: 6 June 2024; Accepted: 16 November 2024;

Published online: 28 November 2024

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Acknowledgements

The authors would like to thank the Long Island University-Dassault Systèmes Center of Excellence for their financial support.

Author contributions

N.R.G. supervised and led the project administration. N.R.G., S.C.D.P., M.C., S.M.L., and J.M.V. were responsible for the project's conceptualization. S.C.D.P. performed the simulation setup, data collection, and curation. N.R.G. and S.C.D.P. analyzed the results and wrote the original draft. V.C. and J.C. helped to implement parts of the computational pipeline and data analysis. D.G., K.E.B., G.K.S., S.A., S.D., and J.M.V. provided important clinical and pathological intellectual content for the study design and results analysis. All authors critically reviewed the paper and contributed important intellectual content. All authors have read and approved the manuscript.

Competing interests

S.M.L. is a Dassault Systèmes employee. J.M.V. is a member of IQDILI and the NIH NIDDK Drug-Induced Liver Injury Network Data Safety and Management Board. N.R.G. has served as a consultant for Dassault Systèmes. The remaining authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Nicolas R. Gallo.

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